

Separating Evidence, Memory, and Programs in Annotation-Policy Adaptation: An Auditable Diagnostic Pilot

Research Working Draft

Authors to be confirmed

Abstract

Can fixed-weight agents learn annotation conventions from a small number of additional examples? We compare access to original examples, persistent notes, and permission to write executable programs under controlled information access. We audit linguistic, political-text, and visual resources, and execute real API-based pilots using two GPT-5.4 model tiers. The SNACS pilot comprises 624 planned prediction episodes over 50 targets in 24 held-out sentences, with additional-label budgets 0, 4, and 16. We also run a real-image Fashion-pedia pilot and 60 non-LLM controls on the full SNACS test projection. The diagnostic results expose substantial threats to interpretation: unequal provider failures, limited trajectories, taxonomy-only visual instructions, and zero program executions in the program-enabled arms. Thus the measured pipeline scores do not establish an advantage of self-modification, a reliable model ranking, or a sufficient number of examples. We provide reproducible inputs, predictions, failure accounting, and a concrete design for the next controlled study.

Diagnostic working draft for review, not submission-ready. Real experiments have run. Provider errors and exhausted API credit prevent a clean, complete comparison. We retain failures and distinguish operational completion from scientific validity.

1 Research Question

An annotator may understand a label definition yet remain uncertain about its application to a particular construction. A motivating observation from Uniform Meaning Representation concerns occupational roles expressed through apposition. This is a hypothesis about an operational gap, not an established omission in the official manual: related role predicates already appear in public documentation, and failure may instead reflect scattered rules

or difficulty applying them.

We study *annotation-policy adaptation*: how a fixed-weight system uses newly revealed annotations to improve decisions under a specified convention. The question is not whether machine learning can read guidelines. Guideline-conditioned extraction already exists, for example in GoLLIE (Sainz et al., 2024). Nor is iterative guideline refinement itself novel. Kim et al. (2026) directly studies guideline reuse, refinement, minimal supervision, and model differences on biomedical named-entity recognition.

We compare systems that retain purchased examples, additionally maintain free-form notes, or additionally permit executable programs. The intended distinction is between access to evidence and persistent adaptation. Our pilots assess implementation and within-sample behavior; they do not yet identify construction transfer, policy-change robustness, or a generally sufficient annotation budget.

2 Related Work and Novelty Boundary

Guideline-conditioned annotation predates this study: GoLLIE uses annotation guidelines for information extraction (Sainz et al., 2024), while GuideBench evaluates domain-oriented rules and updates (Diao et al., 2025).

The closest work is Kim et al. (2026), whose full text and released implementation we inspected. Their moderation procedure predicts on an adaptation set, groups prediction–gold discrepancies, contrasts errors with verified correct cases, generates a principle with negative constraints, and revises the complete guideline. Updates are retained only when adaptation-set strict-match F1 improves, with an additional stopping threshold of 0.9. Each biomedical task uses ten training documents for refinement and 100 development documents for independent evaluation. Persistent guideline refine-

ment, few-document adaptation, held-out evaluation, model comparisons, and refinement-induced regressions are therefore established precedents.

Their ten-document setting does not establish a sample-efficiency curve; varying the refinement-data amount is explicitly future work. Our intended extension compares persistent representations under matched access to examples and varying budgets, with retrieval and execution diagnostics. Free-form notes are not a reproduction of their structured moderation method. Their public NCBI inputs and code are available locally, but a faithful model-based reproduction remains unfinished; no novelty or superiority claim rests on its absence.

3 Cross-Domain Benchmark Audit

Linguistic annotation. UMR offers semantic graphs; SNACS distinguishes scene role from lexical function; GUM/DISRPT offers discourse relations; UD EWT offers candidate natural policy revisions. These remain concentrated in linguistic annotation. STREUSLE and UD EWT share review texts, so they cannot count as independent corpora. Relation classification with supplied discourse units is not full discourse parsing.

Official resources expose two evaluator hazards. UD LAS ignores relation subtypes, requiring full-label or phenomenon-specific measures for changes such as `nmod:desc`. DISRPT files may include an original fine-grained gold label as well as the target label; deleting only the final column does not remove the answer.

Manifesto. Manifesto extends the scope to political-text coding (Manifesto Project, 2026). The required resource is the coded quasi-sentence Corpus, not party-election aggregate statistics. We propose a first track with supplied official segmentation, access to unlabeled document context, one handbook version and one language. Documents, translations and near duplicates must be grouped across splits. Political knowledge, segmentation, country, election year and handbook changes cannot all vary and then be interpreted as a single policy-learning effect.

The official client distinguishes condensed codebook descriptions from the complete handbook. Its v5-to-v4 mapping contains exceptions, so truncating decimal codes is incorrect. Neither the inspected official repository nor the current CRAN package bundles a substitute sentence-level corpus. Corpus access requires an API key or legitimate

export. We acquired the 2021 and 2026 v5 handbooks and terms, but not the licensed sentence-level corpus, which requires account access. The handbook permits progressively wider context up to the whole document and political background; a fixed short window would be an ablation, not faithful implementation. Terms restrict redistribution.

Fashionpedia. Fashionpedia supplies localized garment categories and attributes (Jia et al., 2020). We propose a given-region category/attribute track, separating end-to-end segmentation. Supplied regions reduce localization demands but do not remove perception and fashion knowledge as confounds.

Official documentation lists 46 categories and 294 attributes. The API’s two-image test sample has 48 categories and 320 attributes and explicitly warns that it is not final data. We do not use it for empirical claims. Taxonomy names and hierarchy are not a complete annotation manual; absent attribute entries are not automatically evidence of visual absence.

Official joint AP combines localization and attribute criteria with a specified attribute-F1 implementation. Our proposed supplied-region accuracy and multilabel F1 would be research diagnostics, not official joint AP. Annotation/ontology licenses must be distinguished from source-specific image rights. We acquired the official validation annotations and a bounded set of genuine train/validation images. The visual pilot below uses these images, never the API demonstration sample.

4 Evaluation Protocol

4.1 Information and State

The trusted controller owns raw gold, budgets and scoring. Models access versioned guideline fragments and purchased examples through bounded tools. They may persist notes; a separate condition may write standard-library Python programs. They cannot change the evaluator or purchase labels without recording them.

Supervision is measured as revealed examples, with additional counts of labeled decisions and guideline-embedded examples. Repeated reads consume compute but do not increase the unique-example count. One SNACS example is a sentence revealing all retained target labels. Gold target spans are additional structural supervision shared by all conditions.

At evaluation, every instance starts with the same frozen asset snapshot. Programs execute with read-only assets, cleared environment, isolated process/network namespaces, bounded CPU/output and no mounted repository or gold. An actual sentinel-file, environment and network test verified this isolation. There is no unrestricted fallback.

4.2 Conditions and Identification

Paired conditions are frozen retrieval, persistent free-form notes, and notes with permission to write and execute programs. In SNACS, the official guideline remains accessible in every condition; notes do not replace it. Frozen systems retain all the same purchased examples and fixed structural checks. Full-context prompting, when feasible, controls for tool-mediated access. Active selection is a separate experiment: choosing evidence and using evidence answer different questions.

We distinguish output validity, semantic correctness, transfer and cost. A code change that only improves serialization does not demonstrate semantic learning. Oracle relevant evidence diagnoses retrieval failures. Strong/weak-model asset exchange can help distinguish asset construction from execution, with teacher costs counted and hidden teacher calls prohibited.

4.3 Sample Efficiency and Generalization

We would predefine a quality target and report the smallest measured additional-label budget reaching it with stated uncertainty. An unreached target is a censored outcome, not permission to extrapolate an exact sample requirement. Learning-curve area uses a shared $\log_2(1 + B)$ interval, defined at zero budget.

Public manuals and examples may occur in pre-training. We measure additional deployment-time supervision, not from-scratch sample complexity. Unseen-construction claims require auditing all authorized examples; a random split only establishes unseen-instance evaluation. Expert-designed new conventions and policy revisions offer diagnostics, not contamination guarantees.

5 Completed Control Experiment

5.1 Data and Setup

We pin STREUSLE ([STREUSLE maintainers, 2025](#)) to commit 8ba61fe4 (full SHA in the repository). Files are downloaded with TLS verification and SHA-256 provenance. We project

official JSON into text, tokens and supplied target spans, removing syntax, MISC and other gold layers from predictor inputs. The scorer holds role/function pairs separately.

The training projection has 1,911 sentences, 4,553 retained target decisions and 347 documents. Development has 267 sentences, 462 targets and 156 documents. Test has 259 sentences, 485 targets and 146 documents. Document sets are disjoint. These counts describe our supplied-target projection, not all STREUSLE lexical annotations. Label validity uses the pinned official ontology rather than test-observed labels.

Budgets are 4, 16, 64 and 128 sentences, with seeds 11, 23, 47, 83 and 101. Each seed defines a shuffled nested training sequence shared across methods. Settings were recorded before generating test predictions. These controls do not access a guideline, invoke an LLM or modify programs.

5.2 Methods and Metrics

Majority predicts the most frequent observed joint role/function pair. **Lexical majority** uses the most frequent pair for the target expression, backing off to global majority. **TF-IDF nearest** marks target tokens, represents target and sentence with word unigram/bigram TF-IDF fit only to purchased inputs, and copies the nearest purchased target’s pair. Zero similarity backs off to majority.

The primary metric is exact joint role/function accuracy over retained targets. We also save marginal accuracy, macro-F1, sentence exact match and ontology-valid output rate. The three marginal/joint accuracies and target denominators match the unmodified pinned official scorer for all 60 valid supplied-target runs; this does not establish equivalence for arbitrary invalid or auto-identification outputs. Gold self-scoring reaches 1.0; empty predictions reach 0.0. Predictors are tested without any test-gold field.

5.3 Results

Table 1 reports all 60 runs without seed selection. At 128 sentences, lexical majority reaches 32.66% versus 25.03% for TF-IDF nearest, a paired difference of 7.63 percentage points. A descriptive two-axis bootstrap over paired trajectories and test documents yields a 95% percentile interval of 3.62–11.48 points (2,000 resamples). Five trajectories provide limited evidence: this is not an agent hypothesis test or a general effect estimate.

Budget	Majority	Lexical	TF-IDF
4	4.70 $\pm$ 6.06	6.52 $\pm$ 6.14	4.08 $\pm$ 0.78
16	5.11 $\pm$ 1.19	11.26 $\pm$ 2.85	11.51 $\pm$ 2.76
64	12.82 $\pm$ 4.98	27.38 $\pm$ 5.00	21.28 $\pm$ 3.18
128	15.05 $\pm$ 0.00	32.66 $\pm$ 3.65	25.03 $\pm$ 1.72

Table 1: Completed non-LLM controls: joint accuracy (%) as mean $\pm$ adaptation-run standard deviation over five paired trajectories. Standard deviations are not confidence intervals.

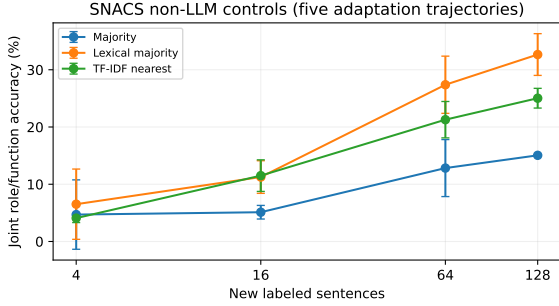


Figure 1: Measured controls. Error bars describe adaptation-trajectory variation; no agent curve is implied.

A sentence is not a fixed amount of supervision: four purchased sentences reveal 6–10 target decisions across these trajectories; 128 reveal 269–308. This motivates multi-unit budget accounting, not a universal sample requirement. The relative strength of lexical majority also motivates retaining lexical controls rather than assuming general sentence-similarity retrieval is sufficient. It does not demonstrate superiority over neural retrieval or parsing.

6 Real Agent Pilot

6.1 Materials and Fixed Pilot Settings

The full SNACS v2.6 manual has 114 pages and 541 uniquely numbered example groups. These are not 541 independently counted labeled targets: subexamples, contrasts and inline examples remain. Budget zero means no newly purchased sentences, not no examples or no prior knowledge. A normalized complete-sentence overlap screen found no long exact corpus-sentence matches, which does not exclude fragments, paraphrases or pretraining exposure.

We use dated API snapshots gpt-5.4-2026-03-05 and gpt-5.4-mini-2026-03-17. Both use native tools with reasoning_effort=none,

Tier	B	Frozen	Notes	+Code
Mini	0	14.0	–	–
Mini	4	25.0	30.0	25.0
Mini	16	19.0	33.0	34.0
Strong	0	24.0	–	–
Strong	4	55.0	31.0	22.0
Strong	16	48.0	27.0	34.0

Table 2: Observed SNACS *pipeline* joint accuracy (%). Every planned target remains in the denominator; failed episodes predict nothing. B4/B16 average two trajectories. Unequal HTTP failures confound method and model comparisons. “+Code” means permission, not observed execution.

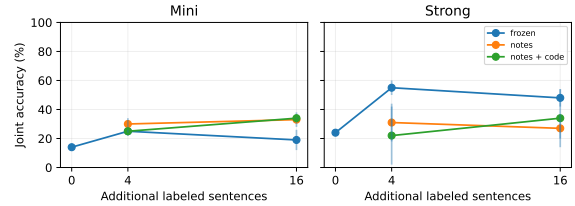


Figure 2: Observed pipeline scores, not identified learning effects. Vertical spans show the two trajectory values. Provider failures are included as errors; the common B0 applies only to frozen retrieval.

the supported setting of the tested transport; this is not a comparison of maximal reasoning capability or known parameter counts. Guides are searched in overlapping text windows with word-boundary and PDF small-caps normalization. Purchased examples are ranked by lexical overlap over text and target expressions, without evaluator gold. Every condition can read any purchased example in full.

Before agent test evaluation, we fixed seeds 11 and 23, nested budgets 4 and 16, and a shared random selection of 24 official test sentences (50 targets, 22 documents; selection seed 20261006). A single frozen budget-zero cell is shared as the no-new-label starting point. There are 13 cells per model, hence 312 evaluation episodes per model. Adaptation permits 24 calls and prediction eight, with a 2,000-token completion ceiling. Asset directories are listed initially; contents require tools. A separate B16/seed11 static control includes the complete guideline and every purchased example in the prompt. Adaptation compute is additional, not compute-matched.

6.2 Pipeline Scores and Failure Accounting

For Mini, 275/312 prediction episodes completed, 20 failed with HTTP errors, and 17 exhausted the

tool-call limit. For the larger tier, these counts are 206, 95, and 11. All eight Mini adaptation episodes completed; only five of eight larger-tier adaptations completed. The original runner recorded HTTP exception classes without machine codes. Later diagnostic replays showed both a recoverable 401 and a 429 with `credit_balance_exhausted`; we cannot retrospectively assign every earlier failure to either cause. New calls were stopped, and the runner now records safe machine codes and halts immediately on exhausted credit.

Mini B16 notes and program-enabled means are 33% and 34%, versus 19% frozen retrieval. The exploratory paired document/trajectory bootstrap interval for notes minus frozen is $[-5.0, 29.4]$ percentage points; two adaptation draws are insufficient for stable population inference, and the interval does not account for availability bias. The Mini full-context B16 control reaches 34% with 70.8% valid outputs in one trajectory. This single-trajectory control supplies a competing explanation, without establishing equivalence or explaining the entire notes–retrieval difference. The larger-tier full-context control stopped before its cell completed and is excluded from aggregate comparisons, not scored as a completed zero.

Program use is a failed manipulation. Neither model called the program-execution tool during any formal SNACS adaptation or prediction episode. The arm therefore tests program permission under this controller, not the effect of learned executable annotation logic. We cannot interpret its table column as evidence for or against program-based self-improvement. Notes were actually read in evaluation, but access alone does not identify their causal benefit.

6.3 Fashionpedia Engineering Pilot

We obtained 16 genuine official-training images through bounded range requests and selected 16 validation images with seed 20261006. Training images come from an annotation-file prefix, not a representative random sample. For each image, a fixed image-specific random seed selects one valid bounding box without inspecting its class or attributes. Crops include 10% context relative to the longer box side, a red box outline, and aspect-preserving resizing to at most 512 pixels. Models see 46 category names and a searchable inventory of 294 attribute names and hierarchy; this is explicitly not a complete annotation manual.

Mini completes the five planned configuration

loops: frozen B0/B4/B16 and notes B4/B16, using one nested trajectory and the same 16 validation regions. Category accuracies are respectively 31.25, 37.50, 12.50, 18.75, and 12.50%. Published-attribute micro-F1 is zero in every cell: the 16 targets have 49 recorded attributes, and no predicted attribute ID matches. Three targets have empty recorded attribute sets. Empty/empty instances add no true positives; unlisted attributes are not certified semantic negatives. The metric is annotation agreement, not official joint segmentation AP.

HTTP failures and tool exhaustion remain in these denominators. Mini produces 16/15/8/13/13 completed predictions in the five cells, respectively. Its B16 notes asset is identical to B4: the agent reads additional examples but writes no updated note. Thus B16 is permission to update, not evidence that further refinement occurred. Official attribute IDs are sparse (294 labels, maximum ID 340); validity uses inventory membership, not a contiguous integer range. The larger-tier visual run was interrupted and does not provide a valid model comparison. With one region per image, prefix-selected training data, incomplete operational instructions and only 16 validation regions, this experiment validates an image/tool pipeline. It does not show a visual guideline-learning boundary or general model ranking.

7 Interpretation and Next Decisive Experiment

The observed curves are nonmonotonic, but this cannot be assigned to overfitting: retrieval, tool limits, incomplete adaptation, and provider availability also vary. A general answer to “how many examples are sufficient?” requires a preregistered quality target, a wider budget range, more trajectories and stable evaluation. No such sufficient sample count is established here.

Three AI reviewer agents independently examined contribution, validity and resources; they are not human annotators or conference reviewers. Their critiques led to native tool calls, better retrieval, explicit asset discovery, file-locked cost accounting, and complete failure denominators. Fifteen software checks now pass, alongside the previously verified real program isolation test. Test fixtures never supply empirical model scores.

The next decisive study must (i) reproduce structured guideline moderation and retain strong full-context/retrieval controls, (ii) complete stable

paired learning curves on adequate held-out data, and (iii) verify at least one expert-defined convention or policy shift, then separate retrieval, rule induction, perception and execution failures. These are prerequisites for an ACL contribution, not a claim that the current sample draft meets them.

8 Limitations

This draft contains genuine controls and diagnostic agent runs, but the requested full research study remains incomplete. Twenty-four SNACS sentences and two trajectories are low power; fifty targets are not fifty independent documents. HTTP failures are unequal across arms and confounded with run order. Availability-adjusted or successful-only comparisons would introduce additional selection bias, so we do not present them as model capability estimates. Failed adaptations can leave partial assets, as registered.

There is no human sample-efficiency baseline, expert-confirmed guideline gap, policy-revision experiment, or audited unseen-construction split. The motivating UMR apposition case is not empirically established here. Public pretraining contamination is unknown. Supplied linguistic spans and visual boxes remove structural prediction, and visual crops retain perception confounds. Manifesto has no empirical results. The absence of executed programs prevents conclusions about program self-evolution. A controlled, fully funded rerun and the closest-method comparison are still needed before submission.

Ethical and Reproducibility Considerations

We retain source hashes, split IDs, predictions and settings while avoiding redistribution of raw licensed texts or images. Credentials are not stored in source, results or manuscript. Dataset and software licenses are distinguished. Political-text gold is not treated as a universal definition of political meaning. Later human studies require appropriate consent, compensation and review. AI assistance in design, software and this draft is disclosed; expert adjudication has not been simulated.

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